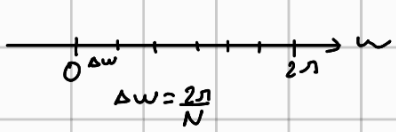
DTFT

$$x[n] \rightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

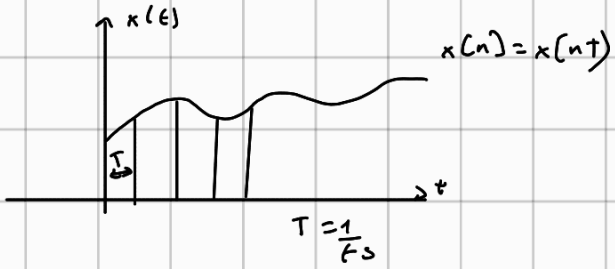


$\Delta\omega = \frac{2\pi}{N}$

DFT

ω örneklendi

$$x[n] \longleftrightarrow X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\left(\frac{2\pi}{N}k\right)n}$$

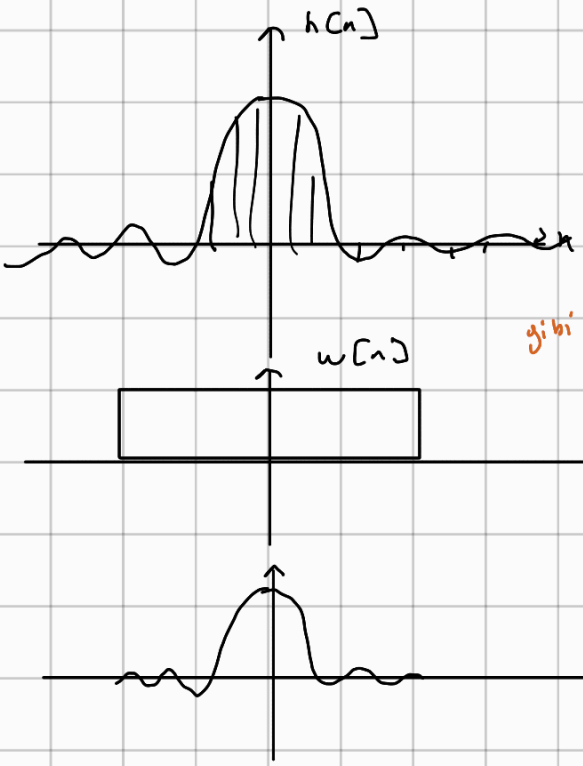




$X[k] = X(k \cdot \Delta\omega)$

$= X\left(\frac{2\pi}{N} \cdot k\right)$

$k = 3 \quad \left(\frac{2\pi}{N} \cdot 3\right)$



gibi

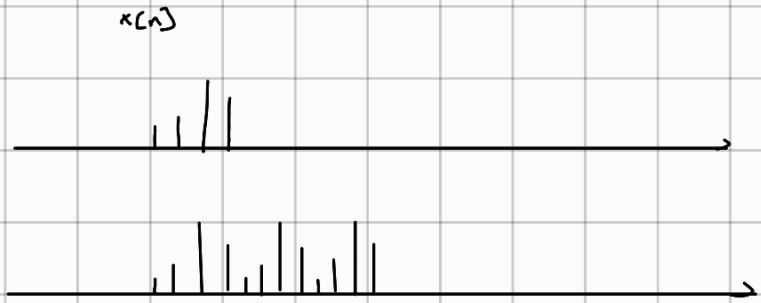
yenibir konu sonra ilişkilendireceğiz

Periodic sequences

periyodik ise $\tilde{x}[n] = \tilde{x}[n+N]$ tüm n'ler için sağlanıyorsa periyodiktir

Discrete fourier series

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n]\, e^{-j\frac{2\pi}{N}kn}$$



$$\tilde{x}[n] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{X}[n]\, e^{j\frac{2\pi}{N}kn}$$

Periodic Convolution

$\tilde{x}_1[n], \tilde{x}_2[n]$ with period N

DFS $\rightarrow \tilde{X}_1[k], \tilde{X}_2[k]$

$\tilde{X}_3[k] = \tilde{X}_1[k]\, \tilde{X}_2[k]$

$$\tilde{x}_3[n] = \sum_{m=0}^{N-1} \tilde{x}_1[m]\, \tilde{x}_2[n-m]$$

periodic convolution with period N (Dairesel konvülasyon)

Linear Convulation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

x_3 ← r_n ← x_2

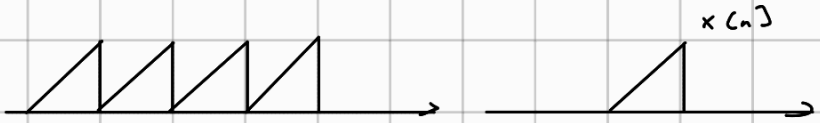
$$x_3[n] = \sum_{m=-\infty}^{\infty} x_1[m]\, x_2[n-m]$$

$$\tilde{X}_3[k] = DFS[\tilde{x}_1[n]] \cdot DFS[\tilde{x}_2[n]]$$

$$\tilde{X}_3[k] = \tilde{X}_1[k] \cdot \tilde{X}_2[k]$$

$$x_3[n] = IFT\left(DFT(x_1[n])\,DFT(x_2[n])\right)$$

(DFT: bilgisayarda, IFT: bilgisayarda)



Bun DFS'i ile bunun DFT si arasında her hangi bir fark yok

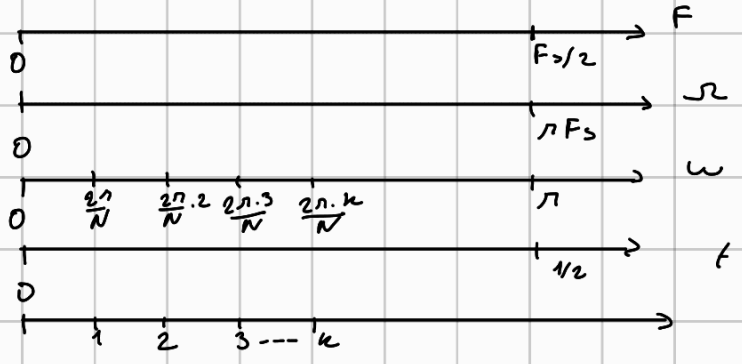
DFT'nin formülü mecburen geldi o mecburiyet bizi DFS'e gönderdi.

Çıkan sonuç yine dairesel konvülasyon sonucudur

$$\sin(2\pi 5t) = \sin(10\pi t)$$

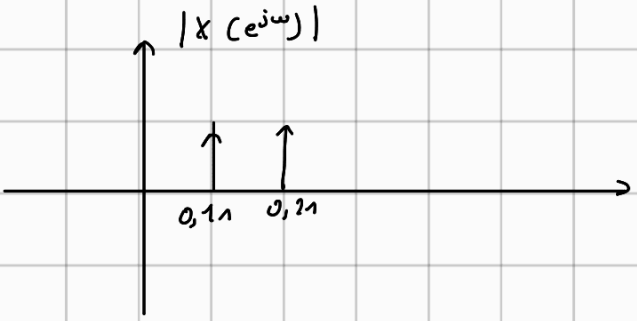
$F_s = 100Hz$

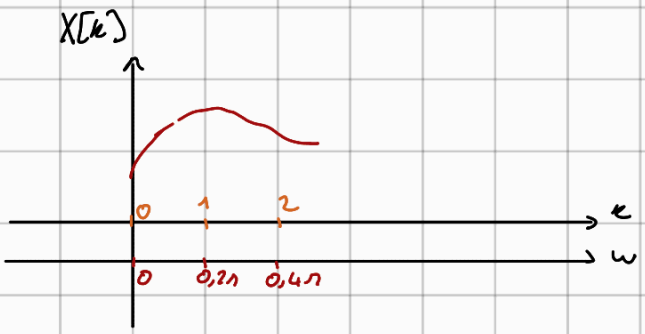
$$x[n] = \sin\left(\frac{\pi}{10} n\right)$$



$$\omega_k = \frac{2\pi}{N} k \qquad x[n] = \sin\left(\frac{\pi}{10} n\right)$$

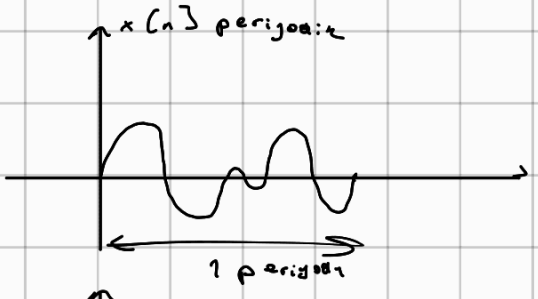
$x[n] = \cos(0,1\pi n) + \cos(0,2\pi n)$



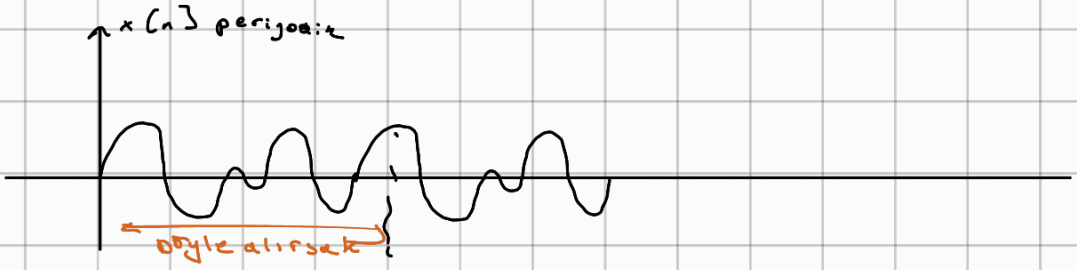


$N=10 \Rightarrow \Delta\omega = \frac{2\pi}{10} = 0,2\pi$

$N=20 \rightarrow \Delta\omega = \frac{2\pi}{20} = 0,1\pi$

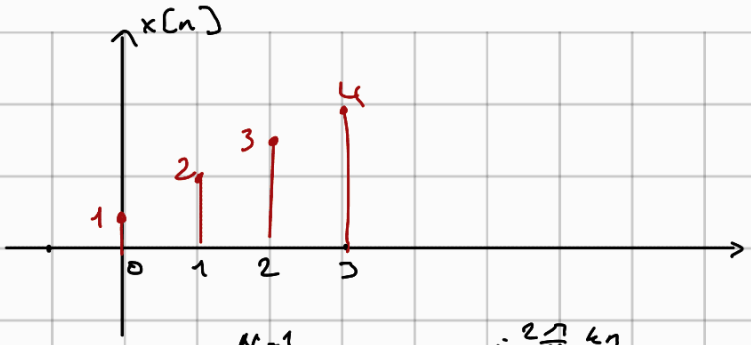






Bu işaretin Fourieri alınmış olur



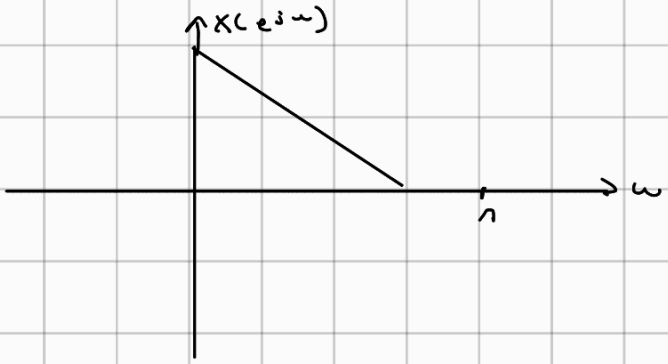


$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}$$

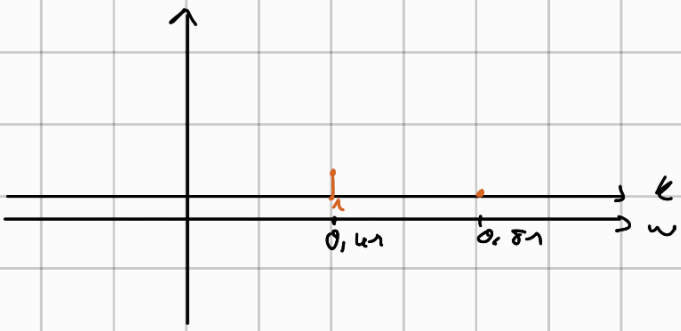
$$X[k] = \sum_{n=0}^{3} x[n]\, e^{-j\frac{2\pi}{4}kn} \qquad k = 0, 1, 2, 3$$

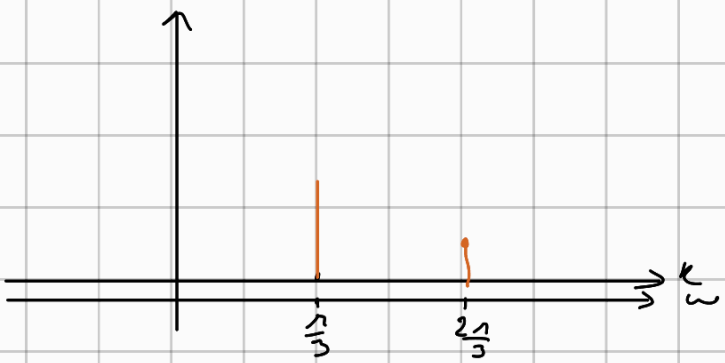
$$X[k] = \sum_{n=0}^{4} x[n]\, e^{-j\frac{2\pi}{5}kn} \qquad k = 0, 1, 2, 3, 4$$

$$X[k] = \sum_{n=0}^{5} x[n]\, e^{-j\frac{2\pi}{6}kn} \qquad k = 0, 1, 2, 3, 4, 5$$









Hesap gücünü azaltmalıyım

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn} \qquad k = 0, \dots, N-1$$

$e^{-j\frac{2\pi}{N}} = W_N$

$$= \sum_{n=0}^{N-1} x[n]\, W_N^{kn}$$

$W_N = e^{-j\frac{2\pi}{N}}$

$y = x * h$

$$y[n] = \sum_{k=0} x[k]\, h[n-k]$$

N tane

$x[n]$ complex $\Rightarrow$ N^2 complex mulG.

$N(N-1)$ summation

for H N^2 complex multiplication

$y = x\,H \rightarrow N^1$

↳ y $\qquad N^2 + N^2 + N + N^2 = 3N^2 + N$

belli koşullar altında Fourier dönüşümü ilişkisi bulmak, daha verimli olabilir

$N = 2^v$

	$x[0]$	$x[1]$	$x[2]$	———	$x[N-1]$
even	$x[0]$	$x[2]$	$x[4]$	———	$x[N-2]$
odd	$x[1]$	$x[3]$	$x[5]$	———	$x[N-1]$

$$X[k] = \sum_{n=0}^{N-1} x(n)\, W_k^{kn} \qquad k = 0, \dots, N-1$$

$$= \sum_{\substack{n=0 \\ n \text{ even}}}^{N-2} x[n]\, W_k^{kn} + \sum_{\substack{n=1 \\ n \text{ odd}}}^{N-1} x[n]\, W_N^{kn}$$

n even için $n = 2r$

n odd için $n = 2r+1$

$n = 2r+1 \Rightarrow r = \frac{n-1}{2}$

$n = 1 \rightarrow r = 0$

$n = N-1 \Rightarrow r = \frac{N}{2} - 1$

$$= \sum_{r=0}^{\frac{N}{2}-1} x[2r]\, \underbrace{W_N^{2rk}}_{W_{N/2}^{rk}} + \sum_{r=0}^{N/2-1} x[2r+1]\, \underbrace{W_N^{(2r+1)k}}_{\frac{W_N^{2rk}}{W_{N/2}^{rk}}\, W_N^k} \qquad k = 0, 1, 2, \dots$$

$W_N = e^{-j\frac{2\pi}{N}}$

$$[W_N]^{2rk} = \left(e^{-j\frac{2\pi}{N}}\right)^{2rk} = \underbrace{e^{-j\frac{2\pi}{N/2}rk}}_{W_{N/2}} = W_{N/2}^{rk}$$

$$= \underbrace{\sum_{r=0}^{\frac{N}{2}-1} x[2r]\, W_{N/2}^{rk}}_{\frac{N}{2} \text{ noktalı DFT}} + W_N^k \underbrace{\sum_{r=0}^{\frac{N}{2}-1} x[2r+1]\, W_{N/2}^{rk}}_{\frac{N}{2} \text{ point DFT}}$$

$\frac{N}{2}\frac{N}{2} + \frac{N}{2}\frac{N}{2} = N^2/2$

FFT

$O(N \log_2 N)$

N^2

N	N^2	$N \log_2 N$
4	16	8
10	256	64
1024	1E6	10240